

EFFECT OF PRUNUS SPINOSA AND AMYGDALUS EXTRACTS ON STRESS INDUCED CHANGES IN MICE

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ABSTRACT:

Aqueous and oil extracts of two prunus species namely *Prunus spinosa* and *Prunus amygdalus*, were evaluated for their ability to reduce the induced stress changes in mice. Plasma corticosterone (PCS), glucose and blood cholesterol levels, were affected by the intake of aqueous extract of *Prunus spinosa* but no significant change was observed with oil extract of *Prunus amygdalus* in mice under stress.

INTRODUCTION

The two species of *Prunus* (Roseaceae) i.e. *P. spinosa* and *P. Amygdalus* commonly known as black thorn and almond respectively, are often used in folk medicine to cure different diseases. *P. Spinosa* has been reported as aspirant, astringent, diaphoretic, diuretic, stomachache, reliable purgative. They are usually used in mild bladder and skin problems, cataract, stomach cramps, dropsy, inflammation in mouth and throat and as antidepressant (Otano et al 1991, Lust 1994). The oil of *P. Amygdalus* is regarded as demulcent, emmollient and pectoral while its bitter variety possess sedative property (Nad Karnio Chopra 1991, Lust 1994). The aqueous extract of *P. spinosa* and oil extract of *P. amygdalus* were evaluated in present study for their ability to reduce stress in mice.

MATERIALS

1 Kg fresh fruit of *P. spinosa* was collected from Rhode Island, USA and 50 ml

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pure oil of *P. amygdalus* was obtained from the market. NaCl AnalR grade of E. Merck and Tween 80 of Sigma, were used in experiments.

METHODS

Fresh fruits of *P. spinosa* were heated in 400 ml of water at 100°C for 15 minutes and the extract was filtered and lyophilized. 150g of it was dissolved in 100 ml of saline for further experiments. Freshly extracted oil of *P. amygdalus* (50 ml) was dissolved in 0.5 ml of tween 80.

Stress was induced in rats by forced swimming method (Irwin 1972) in 15 mice, divided into three groups. A dose of 750 mg/kg of *P. spinosa* and 0.2 ml of *P. amygdalus* each, was injected subcutaneously in two groups of 5 mice each after inducing stress. The third group of 5 mice was treated as control after inducing stress.

Stress affected parameters were estimated in extract treated and control mice to evaluate the effect of both extracts. Glucose level was estimated by O-toluidine method (Varely 1988), plasma corticosterone (PCS) level by spectrolurimetric method (James et al., 1991) and cholesterol level by simple UV method.

RESULTS AND DISCUSSION

In stressed mice, the glucose level increases to significant level as compared to normal mice. When the *P. spinosa* extract was injected, the glucose level decreased to some extent whereas *P. amygdalus* showed no effect on increased glucose level (Table 1).

Cholesterol level lowered in stressed animals as compared to normal but the injection of both extracts did not show any significant effect to bring it to the normal levels. A very slight increase in cholesterol level by *P. amygdalus* was considered insignificant (Table 2).

Plasma corticosterone level used as stress marker, increased in stress conditions but restored to normal after the injection of *P. spinosa* extract whereas the extract of *P. amygdalus* showed no significant effect (Table 3).

The results achieved clearly show that the extract of *P. amygdalus* failed to affect blood glucose, cholesterol and PCS levels (Table 1, 2 & 3), in stressed induced mice. It may be concluded from the results that *P. amygdalus* extract is not good for reducing stress conditions. It may be even harmful due to its high contents of fats and other compounds and therefore should be avoided by the diabetic, hyper cholesterolemic

and hypersensitive patients as it can be dangerous in certain clinical conditions like cardiovascular and blood disorders.

In case of *P. spinosa* extract a slight positive effect on glucose level has been observed in bringing it to normal (Table 1). This may be due to increase in glycogenolysis by stimulation of phosphorylase and/or increased adrenaline leading to hyperglycemia. *P. spinosa* also showed a little effect on cholesterol level (Table 2), but showed a relatively much significant effect on PCS level in keeping it normal (Table 3), which is the most important parameter and used as stress marker. The reduction of PCS level may be due to direct or indirect adrenergic stimulation (Fielding & Lal 1979). Therefore we may conclude that *P. spinosa* extract is helpful in reducing stress effects but we should avoid the use of *P. amygdalus* in our diet as it has showed significant effect on stress and is rather harmful as it changes steroidogenesis in the body.

Table 1
Effect of Prunus Extracts on Glucose Level in Stressed Mice

Condition	Glucose mg/dl
Normal (without stress)	79.9 $\pm$ 2.1
Stressed (Without Extract)	156.0 $\pm$ 1.6
Stressed + <i>P. spinosa</i> (water extract)	100.0 $\pm$ 2.4
Stressed + <i>P. amygdalus</i> (oil extract)	153.1 $\pm$ 2.2

Table 2
Effect of Prunus Extracts on Cholesterol Level in Stressed Mice

Condition	Cholesterol mg/dl
Normal (without stress)	65.8 $\pm$ 3.1
Stressed (Without Extract)	83.0 $\pm$ 2.8
Stressed + <i>P. spinosa</i> (water extract)	70.2 $\pm$ 23.2
Stressed + <i>P. amygdalus</i> (oil extract)	98.5 $\pm$ 3.2

Table 3
Effect of Prunus Extracts on Plasma/Corticosterone (PCS)
Level in Stressed Mice

Condition	PCS Level $\mu\text{g}/100\text{ ml}$
Normal (without stress)	20.7 $\pm$ 1.9
Stressed (Without Extract)	72.8 $\pm$ 2.5
Stressed + <i>P. spinosa</i> (water extract)	20.2 $\pm$ 2.3
Stressed + <i>P. amygdalus</i> (oil extract)	78.2 $\pm$ 2.0

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